

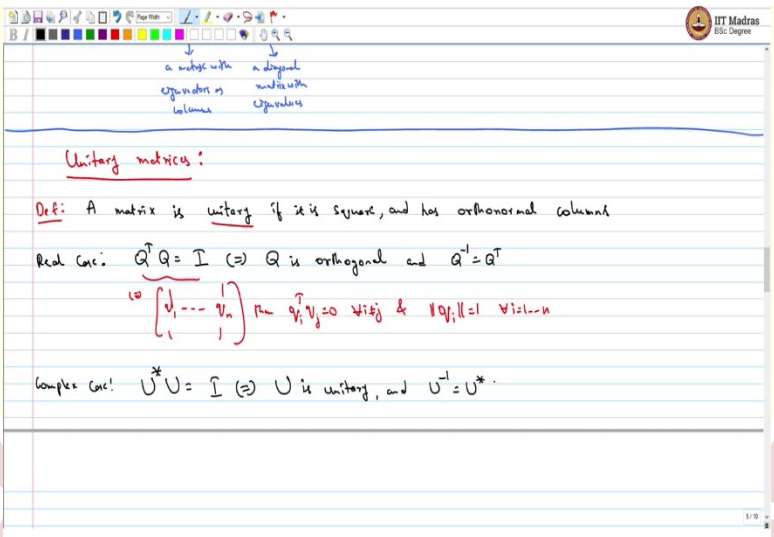
IIT Madras

ONLINE DEGREE

Machine Learning Foundation
Professor Prashanth L. A.
Department of Computer Science and Engineering
Indian Institute of Technology, Madras
Unitary Matrices

Welcome to lecture 3 of Machine Learning Foundations. This is week 5, and in this lecture we will introduce the concept of a unitary matrix, and these matrices, unitary matrices are very important in the context of diagonalization of a Hermitian matrix. These are, you could see these unitary matrices as a natural counterpart of what we have seen before, the so-called orthogonal matrices in the context of a real vector space. Now, we will try to, we will define what a unitary matrix is, and understand some important properties concerning the eigen values and eigen vectors of a Unitary matrix in this lecture.

(Refer Slide Time: 00:54)



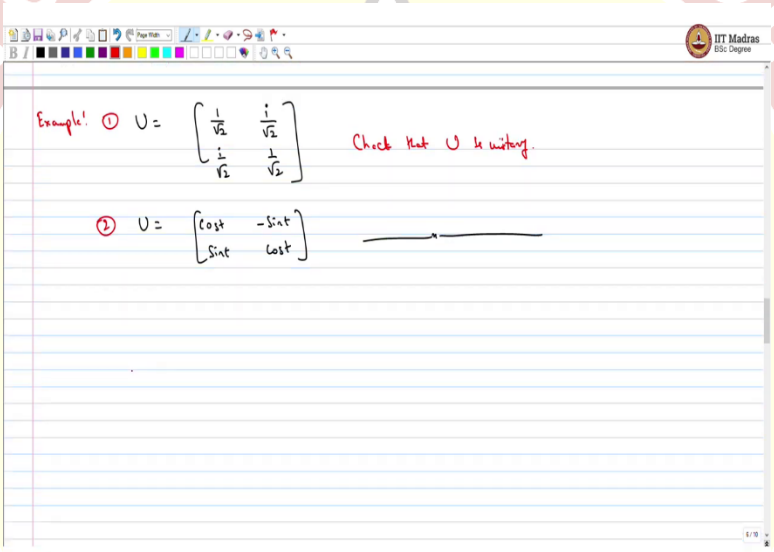
But before that we have to look at a very important, another important class of matrices called unitary matrices. These are the equivalent of orthogonal matrices that we have seen before. So, the definition, a matrix is unitary if, it has to be a square matrix, but that is not enough, and has orthonormal columns.

Now, if you go back to the real case, so what we had was, so our matrix satisfying something like this, we called a orthogonal matrix, it is equivalent to saying Q is orthogonal, which is like saying that if I take any two columns of Q and take their dot product, then it is 0. And the length of each column is 1.So, this is like saying, equivalent

to saying the Q is say, q1 until qn, then qi transpose j j = 0 for all i ≠ j, and the length of this is 1, for all i equal 1 to n. So, that is orthogonality.

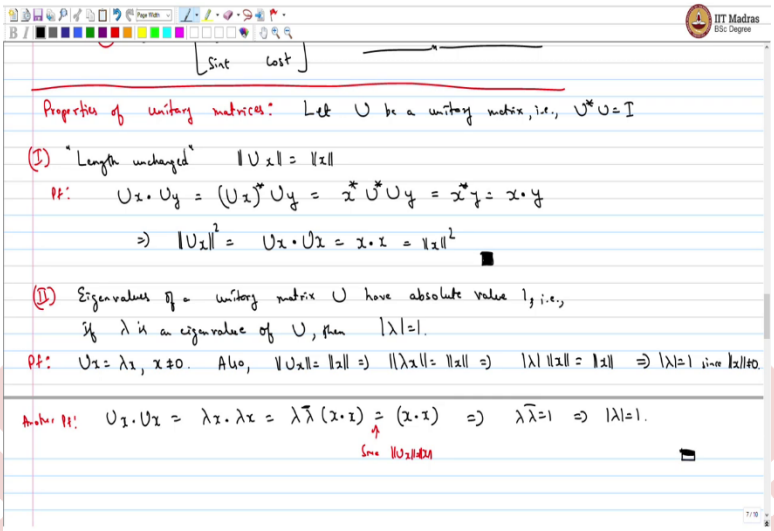
Now, the complex counterpart of this, in the complex case, the counterpart is a unitary matrix, let us denote it by U and instead of just the transpose you have to take conjugate transpose and still recover the same thing. This implies U is unitary. Now, Q transpose, and if Q transpose Q is identity, then that just means Q inverse = Q transpose, the same thing, and U inverse = U *. So, that is a unitary matrix, these are just the complex counterparts or generalization of orthogonal matrix that we have seen before.

(Refer Slide Time: 03:57)



Let us look at an example of a unitary matrix. Do not confuse unitary matrices with Hermitian matrices. These are two different concepts. As homework I will leave this, check that U is unitary. For another example, if you want, take U = cos t, sine t, sine t, this is cos, and cos t. Again, check the same. Check if this matrix is also unitary. Now, as in the case of, unitary matrices are required for diagonalization.

(Refer Slide Time: 05:47)



So, let us look at some properties of unitary matrices. Very first property, we will call this as length unchanged, that is, if I take, for this, let, in all of this, let U be a unitary matrix, that is U * U = identity. So, that, take any vector, then if I apply U, then its length is unchanged.

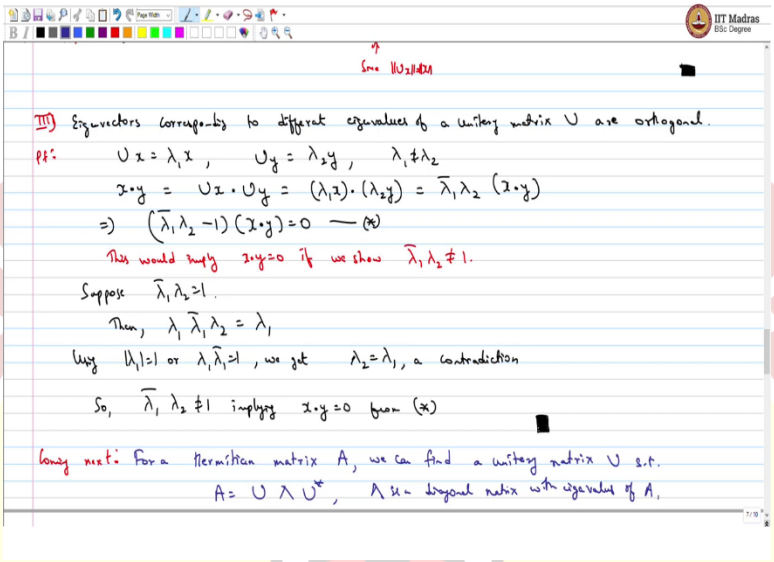
You can show this, so proof, so if I take any two vectors U x dot U y, so this is the same as the conjugate transpose of the first one into second one and U x * is x * U * U y, but U * U is identity. So, this is x * y = x dot y. So, this implies square = Ux dot Ux = x dot x = mod x square. That is the end of the proof. So, unitary matrix does not change the length of any vector that it acts upon. The second property is eigen values of a unitary matrix, U have absolute value 1.

That is, that is, if λ is, and we are not saying the eigen value is real, please note, of U, then we are saying that its absolute value is 1 but not necessarily real. So, λ is not required to be, not necessarily real, and the proof is again showed, so we have U x = λ x. Just to emphasize this is x $\neq$ 0, also norm Ux = norm x, implies, what is Ux, λ x = norm x = norm, absolute value of λ into norm x = norm x.

This just implies λ = 1, since norm x $\neq$ 0. Now, if you want another one-line proof, I can just stream for the same, another proof, I mean the same idea, I am just writing the same thing again. So, you take Ux dot Ux, this is same as λ x dot λ x. Now, this is equal to λ

into, the second λ comes out as λ bar, into x dot x, now is equal to x dot x, since norm U x = norm x. Now, this implies λ into λ bar = 1 or mod λ equal to 1. This is the end of another short proof.

(Refer Slide Time: 11:19)



Now, these two properties are a little different from what we had for Hermitian matrices. Again, a unitary matrix does not change, change the length of a vector, and the second thing is it has, its eigen values have absolute value 1, but a property that it shares with Hermitian matrices is this. It is that eigen vectors corresponding to different eigen values of a unitary matrix are orthogonal. We can show this from first principles. Again, let us do it.

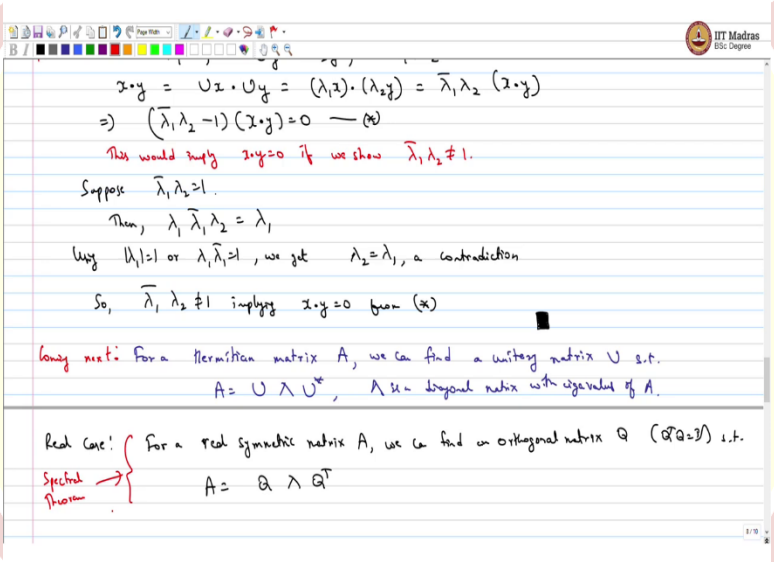
So, we are given Ux = λ 1x, Uy = λ 2y and λ 1 $\neq$ λ 2. Now, we have to look at, what do we need to look at? x dot y. We already know that U does not change the length, so x dot y is the same as Ux dot Uy. But Ux is λ 1x dot λ 2y. Now, this is the same as λ 1 bar λ 2 into x dot y. So, what we have is x dot y on both sides, so I get λ 1 bar λ 2 - 1 $\times$ x dot y = 0. Now, this would imply x dot y = 0 if we show λ 1 bar λ 2 is $\neq$ 1. So, we will do that by contradiction.

Suppose λ 1 bar λ 2 = 1, then get a easy contradiction because λ 1 bar λ 2 into pre-multiplied by λ 1 on both sides = λ 1. Now, using absolute value of λ 1 = 1 from part b, or λ 1 into λ

1 bar = 1, we get λ 2 = λ 1, a contradiction. Let me call this equation *, so λ 1 or λ 2 is $\neq$ 1, implying x dot y = 0 from equation *.

Now, why are we looking at Hermitian matrices, I will just give a preview before ending this lecture. So, coming next for a Hermitian matrix A, we can find a unitary matrix U such that A is U λ U * λ is a diagonal matrix with eigen values of A. So, this is the reason why I introduced unitary matrices. So, we will look at why this decomposition holds. So, this is the classic spectral theorem in some form, for Hermitian matrices. If you specialize this to real symmetric matrices then you get the, then you get the spectral theorem, so well-known spectral theorem.

(Refer Slide Time: 16:33)



This, for the real case, what we have is, for a real symmetric matrix, A, we can find a orthogonal, we can find an orthogonal matrix Q, that is, Q transpose Q is identity, such that A is Q λ Q transpose. This is really the, this is one way of stating the spectral theorem. Multiple ways of stating this, this is one, one way of stating the spectral theorem. So, we will be proving this theorem in sufficient generality for Hermitian matrices and the real symmetric case follows as a easy corollary.

Before I wrap up this lecture I would like to summarize the contents, what did we see in this lecture. Firstly, we saw, we defined what a unitary matrix is. And this matrix will be used for diagonalizing a Hermitian matrix, which is coming next. Now, we saw some two, we saw a few important properties. The intuition is that a unitary matrix, when applied to

a vector does not change the length of the vector. So, that is one thing that we actually showed by using the definition of unitary matrix.

And the second thing we showed about the unitary matrix is that the eigen values of a unitary matrix have absolute value bounded above by 1, which is like saying that they are all within the unit circle, their magnitude. And the last thing is a claim that we had shown earlier also for hermitian matrices, if two eigen values are distinct then the corresponding eigen vectors even of a unitary matrix are orthogonal. Coming next, we will use unitary matrix, matrices to diagonalize a Hermitian matrix. That is in the forthcoming lectures.

